

GAGE, OK, USA

WMO: 746410

Lat: 36.297N

Lon: 99.769W

Elev: 2191

StdP: 13.57

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	13.6	-1.0	5.6	16.3	2.9	6.9	21.4	30.4	50.6	27.7	45.9	8.2	10	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	101.4	70.4	98.8	70.8	96.1	70.7	75.4	91.0	74.3	90.3	73.2	89.3	14.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	126.1	81.2	70.1	120.0	79.9	68.8	115.0	79.0	40.3	90.8	39.2	90.1	38.3	89.4	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	26.8	24.5	-0.3	105.5	5.0	3.1	-4.0	107.7	-6.9	109.5	-9.7	111.3	-13.4	113.5
			-1.1	78.6	4.9	2.5	-4.6	80.4	-7.5	81.9	-10.2	83.3	-13.8	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	36.2	39.1	48.7	57.8	67.5	76.8	81.2	79.9	72.1	59.3	47.1	37.0
	DBStd	18.52	10.20	11.39	11.15	9.74	8.61	6.52	5.59	5.65	7.96	9.31	9.88	9.85
	HDD50	1574	439	332	157	37	2	0	0	0	0	24	167	416
	HDD65	4124	893	725	513	255	75	4	0	0	27	223	539	870
	CDD50	4742	11	28	118	270	545	805	967	928	664	314	81	11
	CDD65	1820	0	1	9	39	153	359	503	464	241	48	3	0
	CDH74	22300	7	36	191	628	1879	4246	6415	5471	2694	640	90	3
	CDH80	11551	1	9	51	231	841	2178	3672	3045	1286	220	17	0
Wind	WSAvg	11.6	10.5	11.2	12.6	13.9	12.8	13.0	11.1	10.2	11.0	11.5	11.2	10.4
Precipitation	PrecAvg	19.2	0.5	0.7	1.5	1.9	3.3	2.8	1.6	2.5	1.7	1.6	1.0	0.7
	PrecMax	28.3	1.4	2.4	5.6	6.0	8.7	7.9	3.7	7.6	5.2	7.9	4.2	2.5
	PrecMin	13.3	0.0	0.0	0.0	0.4	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.4	0.4	0.7	1.4	1.3	2.0	1.7	1.1	1.8	1.2	1.9	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.8	79.3	85.4	92.3	97.9	103.6	105.3	104.6	100.6	91.5	82.4	71.9
		MCWB	50.6	52.2	57.5	60.8	66.2	68.3	70.0	70.4	69.0	63.9	56.7	50.5
	2%	DB	65.9	71.8	79.3	86.6	93.1	98.3	101.5	100.6	96.0	86.0	75.8	64.8
		MCWB	47.0	50.3	56.3	60.1	67.4	70.6	70.9	71.1	69.1	63.7	54.9	48.9
	5%	DB	59.6	64.7	73.8	81.3	89.0	94.6	98.9	97.8	91.6	81.3	70.1	59.5
		MCWB	44.5	48.4	54.6	59.3	66.8	71.0	71.5	71.3	68.9	61.9	53.7	45.4
	10%	DB	54.1	58.6	68.6	76.1	84.3	91.2	95.9	94.5	87.8	76.4	64.5	54.3
		MCWB	42.3	45.1	52.4	57.6	65.4	71.0	71.8	71.2	68.0	60.2	51.6	43.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	57.6	64.1	67.7	73.3	77.3	77.2	76.5	74.1	70.6	62.0	55.9
		MCDB	67.7	70.3	76.3	79.9	89.5	91.0	92.1	92.9	89.7	81.6	69.0	64.1
	2%	WB	48.7	52.8	61.1	65.0	71.0	75.1	75.4	74.7	72.2	67.8	58.8	50.9
		MCDB	63.0	66.2	70.9	76.0	85.5	89.8	91.4	90.9	87.6	77.3	67.8	61.5
	5%	WB	45.7	49.7	58.3	62.6	69.3	73.6	74.3	73.5	70.8	65.6	56.0	47.1
		MCDB	58.6	64.5	68.3	74.4	82.9	89.0	90.7	89.9	85.8	74.2	66.4	58.7
	10%	WB	42.7	46.1	54.3	59.9	67.8	72.3	73.2	72.5	69.5	63.0	53.2	43.7
		MCDB	53.6	57.9	67.2	72.2	79.8	87.0	89.8	88.1	83.6	73.2	63.6	54.0
Mean Daily Temperature Range	5% DB	MDBR	27.2	27.5	28.3	27.3	25.8	24.3	25.0	25.2	25.7	27.2	27.7	25.7
		MCDBR	36.9	37.3	35.8	34.6	30.6	27.9	28.8	29.5	29.1	32.7	35.9	34.6
		MCWBR	21.0	20.1	17.4	15.3	12.4	9.7	8.0	8.7	9.7	14.8	18.5	20.1
	5% WB	MCDBR	34.1	34.4	28.8	27.7	26.5	24.5	24.7	24.7	24.8	24.7	28.5	31.6
		MCWBR	20.1	19.7	15.4	14.8	12.8	10.5	9.0	8.8	9.5	13.3	16.3	19.2
Clear-Sky Solar Irradiance	taub	0.271	0.279	0.317	0.342	0.359	0.397	0.400	0.387	0.355	0.327	0.289	0.273	
	taud	2.551	2.538	2.440	2.399	2.423	2.360	2.378	2.372	2.438	2.498	2.544	2.551	
	Ebn at Noon	299	308	302	298	292	280	279	280	285	285	289	291	
	Edh at Noon	24	28	34	37	37	39	38	38	33	29	24	23	
All-Sky Solar Radiation	RadAvg	910	1146	1519	1876	2095	2293	2249	2012	1700	1267	967	790	
	RadStd	82	136	118	124	184	167	113	107	91	163	106	69	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+169
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page